



Dipartimento di Elettronica e Informazione

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High Performance Processors and Systems

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Name
Last Name

Problem 1 (5 points)	
Problem 2 (5 points)	
Problem 3 (6 points)	
Question 1 (6 points)	
Question 2 (6 points)	
Question 3 (4 points)	
Total (32 points)	

To pass the exam (the following conditions are in AND):

- The overall score has to be equal or higher than 18
- At least 1 problem has to be correct (at least it has to be evaluated with 85% of the “available” points)
- At least 9 points from the problems
- At least 9 points from the questions

Problem 1

Assume that the following code has been executed on a CPU with SCOREBOARD.

	Issue	Read Op	Exec Co.	Write R.
LD F6 32+ R2	1	2	7	8
ADDD F2 F6 F4	2	9	11	12
MULTD F0 F4 F2	3	13	43	44
SUBD F12 F0 F6	4	9	11	12
ADDD F0 F12 F2	5	13	15	16

Is there a "configuration" that can respect the shown execution?
How many units? Which kind? What latency?

Answer

Problem 2

Design and describe (all the decisions have to be motivated) a 1-BHT and a 2-BHT able to execute the following assembly code (R0 is set to 100, R1 is set to 0)

```
LOOP:    LD      F1  0    R0
         ADDD    F2  F1   F1
         ADDI    R1  R1   300
LOOP2:   MULTD   F2  F2   F1
         SUBI    R1  R1   1
         BNEZ    R1  LOOP2
         SUBI    R0  R0   2
         BNEZ    R0  LOOP
```

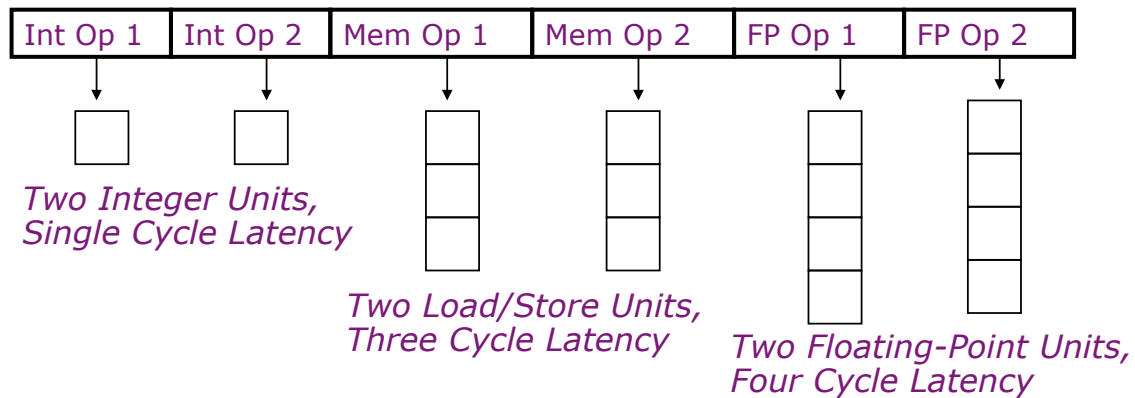
The obtained result, in terms of miss predictions, is inline with theoretical characteristics of the two predictors? Please effectively support your answer.

Answer

Problem 1

P1.A: Explain the main differences between a VLIW architecture and a superscalar architecture, in terms of instructions scheduling, hardware complexity, and code size.

P1.B: Considering the following VLIW "architecture":



Considering the following portion of assembly, describe the corresponding VLIW code:

```
1. Loop:      LD F0, 0(R1)
               FADD F4, F0, F2
               SD F4, 0(R1)
               ADD R1, 8
               BNE R1 R2 Loop
```

How many FLOPS/cycle?

2. With an unroll equal to 4 we can obtain the following code

```
Loop:      LD F0, 0(R1)
            LD F6, 8(R1)
            LD F10, 16(R1)
            LD F14, 24(R1)
            FADD F4, F0, F2
            FADD F8, F6, F2
            FADD F12, F10, F2
            FADD F16, F14, F2
            SD F4, 0(R1)
            SD F8, 8(R1)
            ADD R1, 32
            SD F12, -16(R1)
            BNE R1 R2 Loop
            SD F16, -8(R1)
```

How many FLOPS/cycle?

Answer P3.A

Answer P3.B.1

Int1

Int 2

M1

M2

FP+

FPx

[illegible]

Answer P3.B.2

Int1

Int 2

M1

M2

FP+

FPx

[illegible]

Question 1

1. Explain how Simultaneous Multithreading (SMT) works.

2. Consider a Simultaneous Multithreading machine with limited hardware resources.

Circle the following hardware constraints that can limit the total number of threads that the machine can support and explain why.

For the item(s) that you circle, briefly describe the minimum requirement to support N threads.

(A) Number of Functional Units

(B) Number of Physical Registers

(C) Data Cache Size

(D) Data Cache Associativity

Question 2

Given a cache of a given capacity, associativity and block size answer **TRUE** or **FALSE** to the following questions, motivating your answers.

Doubling the cache capacity of a direct mapped cache usually reduces conflict misses	
Doubling the block size reduces compulsory misses	
Adding a victim cache reduces conflict misses	

Change nesting of loops in the code to access data in order stored in memory will increase spatial locality	
Adding a second level cache reduces the miss rate	

Question 3

- a) Present briefly the register renaming mechanism, explaining how it works
- b) Explain the differences with respect to the Reorder Buffer
- c) Explain how it deals with precise interrupts

